Focus-TransUnet3D: High-Precision Model for 3D Segmentation of Medical Point Targets

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***Abstract*— Deep learning has been extensively applied in medical image segmentation, providing significant support for disease diagnosis. However, traditional encoder-decoder networks struggle with segmenting scale-sensitive point target lesions. To address this challenge, this paper proposes an innovative incremental fusion architecture that can integrate different models and achieve significant performance improvements through complementary fusion. Based on this architecture, we developed Focus-TransUnet3D by combining the Trans-FusionNet3D model and the 3D Unet model. This model adopts a global-to-local segmentation strategy, effectively addressing the challenges of medical point target segmentation, thereby expanding the application of deep learning in the field of medical image processing. Furthermore, we design a deep fusion strategy suitable for the transformer model to adapt to multi-scale feature learning. The integration of the transformer model with convolutional neural networks brings improvements in local and global feature extraction capabilities, enhancing the applicability of our model. We evaluate our model on three clinical datasets with different target scales: the Intracranial Artery dataset, the Intracranial Aneurysm dataset, and the LiTS17 dataset. The results indicate that in the external test for intracranial aneurysm auxiliary diagnosis, the model trained with only 47 annotated samples achieved the state-of-the-art performance, attaining a Dice coefficient of 84.14% and a sensitivity of 100%. This effectively addresses the challenges of annotation scarcity and tiny targets. Our code will be released at https://github.com/caijilia/FTUnet3D.**

***Index Terms*— Medical image 3D segmentation, deep learning, medical point target, intracranial aneurysm, transformer, cascaded network.**

I. Introduction

THE application of deep learning technology in the medical field has achieved great success [1]. Currently, medical image segmentation techniques for imaging modalities such as X-rays [2], CT scans [3], and MRI images [4] have made remarkable strides. Through these techniques, physicians can more accurately locate and analyze lesion areas [5], [6]. However, current research on the precise segmentation of subtle local structural abnormalities is often overlooked. These abnormalities usually represent early manifestations of critical diseases, such as intracranial aneurysms, microbleeds, and other small anatomical anomalies. Accurately identifying these targets is crucial for improving diagnostic accuracy and patient prognosis [6], [7]. Particularly in the field of intracranial aneurysm detection, neurosurgeons' comprehensive analysis of extensive patient scan data confirms that targets of interest for early diagnosis and treatment of intracranial aneurysms typically constitute less than 0.001% of the volume ratio. This characteristic is also indicative of early stages of various other diseases, where the volume ratio of lesion tissue is less than 0.001%, posing a high risk of missed diagnosis. At such a minute scale, traditional morphological features used for differentiation, such as tubular and spherical shapes, lose their discernibility and often exhibit point-like target characteristics [8]. Therefore, we propose a new definition method to classify these targets with a volume ratio of less than 0.001% as medical point targets. Due to specific considerations for tumor treatment applications and medical imaging analysis technology, this definition is different from the broader concept of small targets and most directly reflects the challenges caused by the tiny size of lesions. To achieve accurate identification and segmentation of medical point targets, the direct transfer of existing methods is no longer sufficient. It is necessary to better utilize 3D spatial features to enhance the model's lesion recognition capability.

The original scanning data output from medical imaging devices is often in 3D format, adding a depth dimension compared to natural images. Commonly employed methods involve slicing the original 3D data [9], [10] or utilizing a 2.5D approach to process volumetric data [11]. However, these processing methods cannot fully harness the three-dimensional attributes contained within medical images, including volume, shape, and structure. Hence, for medical point targets like intracranial aneurysms, developing direct 3D segmentation models that can more comprehensively utilize spatial depth information is particularly crucial. However, this endeavor faces several pressing challenges. Firstly, the large scale of 3D data demands higher computational resources and storage. Secondly, data annotation becomes more complex as it requires considering three-dimensional boundaries and

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volumes [12]. Model training and optimization also become more challenging due to the involvement of information propagation and feature extraction along multiple dimensions [5]. Lastly, issues such as noise, deformation, and irregular shapes present in actual medical images make the design of 3D neural networks more complex [13].

Currently, researchers have directly modify classical 2D models by replacing operators such as 3D convolutions to achieve the design of 3D models [14], [15], [16]. However, in medical scenarios with limited annotations, this upscaling risks overfitting. To be more targeted, other researchers have developed specialized 3D models focusing on depth context extraction, somewhat reducing model complexity and parameter scale [17], [18], [19], [20]. But these rely solely on convolutional operations, limiting segmentation accuracy due to finite receptive fields. For comprehensive 3D segmentation of human tissues and organ structures, which often span multiple slices, a larger receptive field is essential. Consequently, achieving precise 3D point-target segmentation remains a formidable task.

To address these challenges, we propose Focus-TransUnet3D (FTU3D), a novel model blending 3D transformer neural network and Convolutional Neural Network (CNN) for high-accuracy point-target lesion segmentation. FTU3D enhances the capability to capture global dependency features, enabling precise differentiation between lesion-specific tissues and normal tissues, thereby improving segmentation accuracy. Within this unified architecture, the 3D transformer model is integrated to utilize slice information across various depths, effectively capturing the overall structural feature distribution. Concurrently, the complementation with the local detail capturing ability of the U-shaped CNN model enables the achievement of refined segmentation results. The main contributions of this paper are as follows:

1) A novel Trans-FusionNet3D model is proposed, which improves the feature embedding method of transformer-based models in visual tasks, thereby increasing the information density of input tokens. It fully leverages cross-spatial information in 3D medical image data to efficiently extract 3D global features.
2) A Hierarchical Feature Fusion module is proposed to achieve feature-level fusion by progressively integrating multi-layer key embedded information from the transformer neural network model. An Incremental Fusion module is proposed to achieve efficient integration of models with different architectures and reduce the interaction cost between models.
3) The Focus-TransUnet3D model is proposed for high-precision segmentation of medical point target lesions. This model efficiently integrates the global feature extraction capabilities of the Trans-FusionNet3D model with the spatial refinement capabilities of the Unet3D model through the Incremental Fusion module. This integration forms an incremental fusion learning architecture, making the training process more efficient and streamlined, and establishing a novel paradigm for multi-model fusion structures.
4) Our model demonstrated state-of-the-art (SOTA) performance in the tasks of intracranial artery segmentation, intracranial aneurysm segmentation, and liver tumor segmentation. This signifies a significant breakthrough in deep learning for auxiliary medical applications, particularly in the field of neurosurgery.

II. Related Work

A. Medical Image Segmentation Based on Deep Learning

The U-shaped Unet [21] architecture has become a standard reference framework in medical image segmentation. Variants like Unet3+ [9] and MSRF-Net [22] have extended its capabilities, exploring complementary effects of feature mappings at different scales and addressing challenges like target variability and global feature extraction. ESNet [23] advances this further by enhancing global perception in CNNs and reducing model parameters. Double-Unet [24] introduces dual U-shaped structures for more profound information extraction. ERDUnet [25] improves this dual structure by combining the efficient module proposed in ESNet to achieve a robust and lightweight model for different medical tasks. In recent times, the transformer architecture [26] has gained traction in medical image segmentation. TransUnet [27] combines transformer architecture with CNN to enrich global context awareness. Transfuse [28] introduces a dual-branch parallel structure of CNN and transformer architecture, effectively incorporating spatially induced bias information. SwinUnet [29] leverages the improved Swin-transformer [30] structure for size change adaptability. SSFormer [31] integrates layered transformers and MLP [32], achieving a lightweight model without introducing segmentation headers.

However, all these methods only design models under a 2D perspective, making them difficult to apply on challenging tasks such as point target segmentation. By directly constructing 3D models, we comprehensively utilize all features of the volume space, which is the basis for successfully overcoming the challenges of medical point target segmentation.

B. Improving Medical Image Segmentation Using 3D Technology

The heightened dimensionality of 3D data imposes rigorous demands on algorithm design. X-CTRSNet [33] constructs 2D kernels that combine slices from different plane projections to approximate 3D segmentation. Hybrid 2D and 3D methods are prevalent [11], [34], [35], extracting 2D features from different perspectives and integrating them with 3D information for segmentation. Some deep learning models attempt to realize 3D segmentation directly based on the framework of the original 2D model through operator replacement [14], [15], [18], [36]. These adaptations show promise in certain datasets but generally suffer from limited generalization, especially in point target segmentation. UNETR [37] and SwinUNETR [38] introduce transformers, using Vision Transformer (ViT) modules [39] and Swin-ViT modules [30] as encoding subnetworks, combined with 3D convolutional modules to formulate a U-shaped segmentation algorithm. TransBTS [40] integrates

ViT modules as transitional elements in a traditional U-shaped architecture.

Although introducing 3D transformer modules can effectively utilize global contextual information for point target recognition, their performance heavily relies on extensive training data. The dense sparse joint training paradigm proposed by DeSCO [41] aids in mitigating the pressure of annotating 3D data but can only leverage information from two orthogonal viewpoints in the original data. Therefore, we propose a novel architecture that uses complementary models to decompose the learning task of transformer-based models, thereby addressing the above problems.

C. Point Targets Segmentation in Medical Field

In the realm of Synthetic Aperture Radar (SAR) image processing, point targets are small, point-like objects or bright spots [8]. In the field of medical imaging, there are many similar tiny targets, such as intracranial aneurysms, tiny abnormalities in blood vessels or other local structures. Accurate identification and segmentation of these targets is critical to improving the quality of medical care. We define targets with a volumetric proportion smaller than 0.001% in 3D scan image data as medical point targets. Currently, the precise segmentation of medical point targets still faces unique challenges. Firstly, their minuscule spatial presence makes them prone to background occlusion, and existing approaches often struggle to effectively capture their features [42]. To address this issue, [43] enhances the recognition of point targets by integrating context features from different time sequences in 2D views. Additionally, noise and artifacts in medical scans often reduce the contrast of point targets, leading to blurred boundaries with the background. To improve this phenomenon, an ensemble learning method is introduced to complete the segmentation task by training three different deep learning models [44]. Furthermore, the deeper network structure and residual connections in DCAU-Net [45] enhance its learning ability for point-target lesions. The transformer architecture also shows promise in capturing contextual information of point targets [7], [46].

Building on these insights, we propose a unified architecture that synergizes different models for enhanced performance. Under this architecture, we design a segmentation network that gradually focuses on point targets from coarse-grained to fine-grained. Crucially, our method demonstrates robust performance across varying scales of segmentation targets. This has general significance for improving patient survival rates and contributing to the advancement of medical intelligence.

III. Methods

A. Network Encoder Based on 3D Transformer

To more fully extract cross-slice global features in 3D medical images, we employ transformer structures to fulfill the encoding function of the network. The inherent quadratic computational complexity of transformers in relation to sequence length, however, makes direct image-to-token conversion impractical. Therefore, we adapt and optimize the vision transformer architecture to suit the nuances of 3D data.

Our optimization primarily focuses on refining the 3D image embedding module to efficiently translate voxel data from medical images into vector representations. We incorporate a pre-encoding step prior to embedding, using max-pooling for feature downsampling. The downsampling rate r, as a hyperparameter, can be adjusted through manual configuration and experimentation. After pre-encoding, we embed image patches via volumetric split to produce a feature map f and unify it to d dimensions ($d = 768$). This approach significantly reduces computational complexity and enhances processing efficiency. It also addresses hardware limitations and mitigates overfitting risks, pivotal in 3D medical image segmentation. It is worth noting that when downsampling results in only one token in the embedding sequence, the model becomes more focused on capturing the internal correlation features of this token. This can naturally cooperate with our incremental fusion architecture, achieving a degradation of the transformer learning task. By effectively capturing and processing 3D image features, our approach promises greater accuracy and reliability for medical imaging applications.

For accurate spatial relationship comprehension in 3D images, a learnable position embedding PE is introduced. This PE is added directly to the feature map f, forming the input feature z_0, as detailed in eq.(2).

$$F_1^{in} = Maxp_r(F^{in}) \tag{1}$$

$$z_0 = f + PE = Embed(F_1^{in}) + PE \tag{2}$$

Among them, $Maxp_r$ represents the max-pooling operation with hyperparameter r, and $Embed$ represents the image embedding operation. F^{in} represents the original input (3D image features). F_1^{in} represents the input features of the Trans-FusionNet3D model. The feature dimension of F_1^{in} is (Batch size, Channels, Depth, Height, Width). The feature dimension of f is (Batch size, Number of patches, Embedding dimension). The relationship between the feature dimensions of these two feature maps is as follows:

$$Number\ of\ patches = \frac{Depth}{16} \times \frac{Height}{16} \times \frac{Width}{16} \tag{3}$$

The transformer encoder in our model consists of L layers, each adhering to a standard architecture encompassing a Multi-Head Self-Attention module (MSA) and a FeedForward Neural Network module (FFN). The utilization of the MSA mechanism enables the model to simultaneously capture diverse relationships in 3D medical images. The standard structure of the l-th layer of the transformer encoder is represented as follows:

$$z_l' = MSA(LN(z_{l-1})) + z_{l-1} \tag{4}$$

$$z_l = FFN(LN(z_l')) + z_l' \tag{5}$$

Among them, LN represents Layer Normalization, and z_l denotes the output of the l-th transformer layer.

B. Trans-FusionNet3D With Deep Fusion Technology

Based on the encoder composed of 3D transformers, we implemented a global granularity model named Trans-FusionNet3D (TFN3D). The hierarchical transformer structure can incrementally deepen the semantic depth of the encoding results, ultimately obtaining the deepest encoded output. We then construct the decoding part using 3D convolutional modules. Specifically, we first use 3D convolutional modules to map the embedded space to the structural space, aiming to reconstruct the 3D spatial structure. Subsequently, we employ trilinear upsampling to gradually increase the volumetric resolution of the data until it is restored to the scale prior to feature embedding, achieving the decoded final output F_1^{de}.

Considering the high-dimensional characteristics and minute scale of 3D medical point targets, we utilized different scale outputs in the decoding part to construct a Hierarchical Feature Fusion (HFF) module. To improve computational efficiency, only the encoded outputs from the 3rd, 6th, and 12th layers are selected as key features, serving as the input sources for the HFF module, corresponding to $F_{1_3}^{en}$, $F_{1_6}^{en}$, and $F_{1_{12}}^{en}$, respectively. We perform different upsampling operations on these three outputs. The first two are uniformly resized to $4 \times 4 \times 4$ before being initially concatenated for fusion. After upsampling the fusion result to a scale of $8 \times 8 \times 8$, a 3D CBAM attention module [47] is introduced to reduce noise. As for $F_{1_{12}}^{en}$, it is directly upsampled to $8 \times 8 \times 8$ for the final fusion, resulting in F_{fuse}^{out}. By applying the HFF module to the decoding part of TFN3D, the previously mentioned final decoded output F_1^{de} is produced, as shown in eqs.(8) and (9).

$$F_{fuse}^{1} = up(Conv(F_{1_3}^{en})) \bigoplus up(Conv(F_{1_6}^{en})) \tag{6}$$

$$F_{1_{12}}^{en'} = up^2(Conv(F_{1_{12}}^{en})) \tag{7}$$

$$F_{fuse}^{out} = Att(up(Conv(F_{fuse}^{1}))) \bigoplus F_{1_{12}}^{en'} \tag{8}$$

$$F_1^{de} = up(F_{1_{12}}^{en'} + F_{fuse}^{out}) \tag{9}$$

Among them, $Conv$ represents a module that includes Convolution3d, InstanceNorm3d, and LeakyReLU. The use of the LeakyReLU function can better handle the diverse characteristics of medical image features, reducing the risk of overfitting by increasing the network's expressive capacity. Additionally, LeakyReLU is computationally simple and highly efficient. up represents the trilinear upsampling method. $\bigoplus$ denotes concatenation. Att indicates the use of the 3D CBAM module as the attention mechanism.

C. Focus-TransUnet3D Under Incremental Fusion Network Architecture

To apply transformer models in medical scenes with extremely few annotations, we developed a 3D segmentation model named Focus-TransUnet3D (FTU3D), through a incremental fusion architecture that integrates the TransFusionNet3D and a CNN model. The specific network architecture is illustrated in Fig. 1.

Before model fusion, we constructed an Incremental Fusion module to reduce the learning cost of information communication between different architectural models. The output feature F_1^{de} from the TFN3D model serves as the input to this module. Additionally, to further compensate for spatial information, we implemented residual fusion using the input feature F_1^{in} of the TFN3D model, as detailed in eq.(12).

$$F_1^{de} = up_{HFF}(F_{1_{12}}^{en}, F_{1_6}^{en}, F_{1_3}^{en}) \tag{10}$$

$$F_1^{res} = LR(Conv(Conv(F_1^{in})) + F_1^{in}) \tag{11}$$

$$F_{IF}^{mask} = Sig(up(Conv(ResC(Att(F_1^{res}) \bigoplus F_1^{de})))) \tag{12}$$

Among them, up_{HFF} represents upsampling achieved by using the HFF module. LR represents the LeakyReLU activation function. $ResC$ implies the use of the residual module in eq.(11). F_{IF}^{mask} represents the output of Incremental Fusion module. Sig represents the Sigmoid activation function.

Through the rich contextual information provided by both residual and hierarchical fusion, our model is capable of learning more robust and universal feature representations [48]. This has significant implications for overcoming the variations between imaging devices and individual patients. The fusion of high-resolution shallow features with deep features enriched with contextual information also enhances the model's ability to recognize tiny structures. The Sigmoid activation function is then applied to obtain mask features for incremental fusion, as detailed in eq.(12). The Sigmoid function can map the features from the TFN3D model into a weight mask for the input image, optimizing feature selection so that the learning focus of the subsequent convolutional model is more concentrated on effective feature areas.

By merging this F_{IF}^{mask} with the original input image features, an incremental fusion connection between the TFN3D model and the subsequent CNN model can be constructed, as detailed in eq.(13). The included residual connection ensure that this incremental fusion primarily focuses on the parts different from the original image information, thereby providing the TFN3D model with clearer training targets and enhanced training signals. This allows the preceding TFN3D model to focus on identifying more valuable regions in the voxel space during training. Consequently, the integrated model can concentrate more on learning lesions in high-value areas, improving the success rate of identifying and segmenting medical point targets.

$$F_2^{in} = F_{IF}^{mask} \cdot F^{in} + F^{in} \tag{13}$$

Among them, F^{in} represents the original input (3D image features).

Subsequently, we constructed the Unet3D model based on the classic Unet2D architecture for the final fine-grained segmentation. As shown in Fig. 1, the encoder subnetwork of the Unet3D model is mainly composed of the CBR module, which contains a 3D convolution function, a Batch Normalization function, and a ReLU activation function. The four-layer feature extraction structure built by the CBR module can provide significant advantages in avoiding gradient vanishing and simplifying calculations. In the decoder subnetwork of Unet3D, the corresponding four-layer feature decoding structure is composed of the CBR module and the 3D transposed convolution function. To perform targeted upsampling, we used multi-scale encoded outputs from the encoding part for skip connections to retain essential high-resolution features. The last layer of Unet3D is constructed as a 3D convolutional output layer with a convolutional kernel of $1 \times 1 \times 1$.

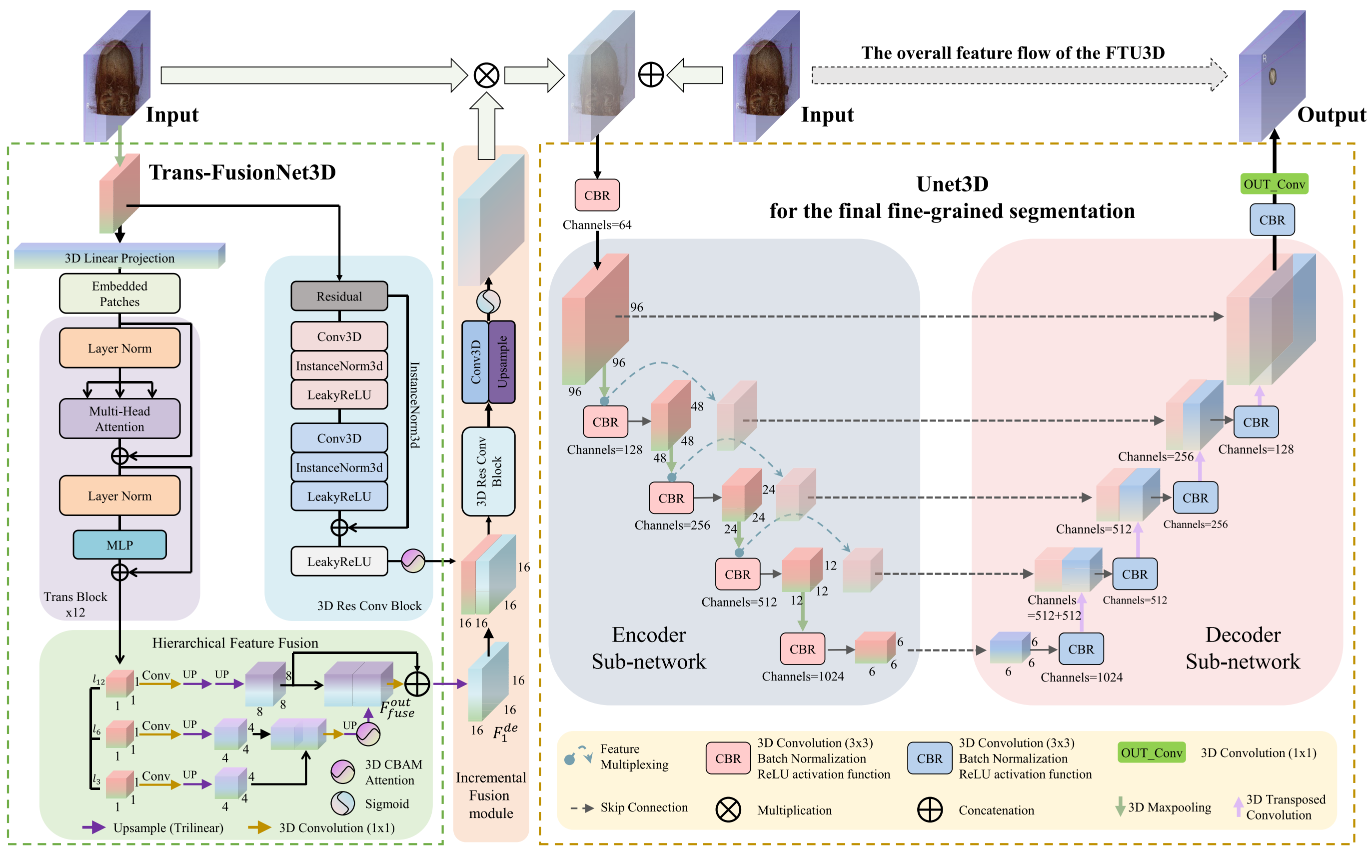


Fig. 1. The network architecture of our proposed Focus-TransUnet3D model. This model is designed as an incremental fusion structure of two fundamental models. The Trans-FusionNet3D model primarily employs a Hierarchical Feature Fusion (HFF) module and a Multi-Head Self-Attention mechanism to extract global features. The Unet3D model is mainly responsible for refining the segmentation results by leveraging the self-repeating structures in medical images to learn spatial details. These two components are integrated through an Incremental Fusion module, which reduces optimization difficulty and enhances training with sparse data.

As seen in Fig. 1, the overall FTU3D model is composed of a cascade of the TFN3D and Unet3D models, leveraging the advantages of both. The TFN3D model excels in global feature extraction, but the complex spatial structure of human tissues limits the effectiveness of the transformer model in the absence of training data. The convolution-based Unet3D can supplement this by fully utilizing the rich self-repeating structures present in 3D medical images to learn finer details. The TFN3D and Unet3D models operate relatively independently in terms of feature learning but establish a communication pathway between the output of the TFN3D model and the input of the Unet3D model through the Incremental Fusion module, as detailed in eqs.(12) and (13). This architecture helps improve the stability of gradients during model fusion, reducing the extra training overhead associated with mutual learning of decoding between different models, making the overall training process more efficient and straightforward. Additionally, it can be used as a regularization mechanism to reduce overfitting and encourage the preceding network to generate more generalized feature representations, ultimately optimizing the learning efficiency and performance of the entire network.

IV. Experiments

A. Datasets

To evaluate the effectiveness of our proposed Focus-TransUnet3D, we validate it on three datasets: the Intracranial Artery dataset (Artery), the Intracranial Aneurysm dataset (IA), and the Liver Tumor Segmentation dataset (LiTS17) [49]. The Artery dataset and IA dataset were derived from actual imaging conducted on patients at Beijing Tiantan Hospital. The patients' ages ranged from 18 to 80 years, with a median age of 50 years, including approximately 50% female and 50% male patients. The Artery dataset includes 21 annotated samples of the entire intracranial artery structure. The IA dataset consists of 47 annotated samples containing regions of intracranial aneurysms. Additionally, to test the generalization performance in practical applications, we conducted external tests for auxiliary diagnosis of intracranial aneurysms using 30 newly acquired samples in a real clinical environment. This External Test dataset includes 15 negative samples and 15 positive samples for intracranial aneurysms.

The data modalities for these datasets are CT angiography (CTA) images, which are high-resolution datasets aimed at advancing lesion visualization and cerebrovascular abnormality detection. The annotations in these datasets are at the voxel level, allowing for high-precision depiction of anatomical structures and pathological features in CTA images. Each image annotation was manually created by medical experts in the relevant field to ensure the highest accuracy and reliability. To further enhance annotation reliability, each annotated image underwent a rigorous validation process involving a second expert review to confirm the accuracy of the initial annotations, thereby minimizing the possibility of errors. Due to privacy concerns, the Artery and IA datasets cannot be open-sourced at present. We have instituted appropriate privacy safeguards in our experimental process to ensure the ethical use of these data without compromising their utility.

The LiTS17 dataset is an open-source dataset aimed at liver and liver tumor segmentation tasks for patients. This dataset comprises CT scan images of 131 liver cancer patients, with voxel-level annotations of liver tumor lesions manually created by medical experts. The dataset was provided for the LiTS2017 challenge, contributed by various clinical sites worldwide. It currently does not include specific demographic information about the patients, such as age, gender, and ethnicity.

B. Evaluation Indicators

We conduct comprehensive internal testing of the model's performance using three metrics: Dice coefficient (Dice), Intersection over Union (IOU), and Average Symmetric Surface Distance (ASSD). Additionally, to ensure the reliability of the internal test results, we employ 7-fold cross-validation. In the external test, given the availability of additional diagnostic results for the condition, we further use Sensitivity and Specificity as evaluation metrics. These two metrics are used to evaluate the model's performance in identifying positive and negative samples, respectively. The specific calculation formulas of these two indicators are shown in eq.(14) and eq.(15), respectively.

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{14}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{15}$$

Among them, TP (True Positive) is the number of positive lesions correctly segmented, and FN (False Negative) is the number of positive lesions that are not correctly segmented. TN (True Negative) is the number of negative lesions correctly segmented, and FP (False Positive) is the number of negative lesions incorrectly segmented.

C. Experimental Setup

All experiments are implemented using PyTorch 1.12.0 framework on a single NVIDIA RTX A4500 GPU. We use weighted Dice loss, as shown in eq.(16). We use AdamW optimizer with a learning rate of 1e-5 to train all models. We set the batch size and epoch to 1 and 3000, respectively. Our choice of network hyperparameters is set to MSA of transformer with 12 attention heads, and the initial downsampling rate r is set to 6. Preprocessing involved uniform resizing of dataset images to a voxel size of $0.449 \times 0.449 \times 0.625$ via nearest-neighbor interpolation, followed by adjustment to a uniform size of $512 \times 512 \times 256$. Random cropping is then performed with a 1:1 ratio of negative to positive samples, resulting in $4 \times 96 \times 96 \times 96$ sub-volumes as input to the model. The voxel intensity range is standardized to -175 to 250. Data augmentation techniques, including random cropping, flipping along three dimensions, 90-degree rotation, brightness transformation, and contrast transformation, are applied to the input during training.

TABLE I
EXPERIMENTAL RESULTS BASED ON ARTERY DATASET

Model	Dice(%)	IOU(%)	ASSD
UNETR [37]	88.15±0.46	80.91±0.55	1.98±0.33
SwinUNETR [38]	89.79±1.37	83.07±2.16	1.93±0.41
ResUnet [36]	89.40±1.23	82.53±1.98	1.74±0.20
Unet [14]	86.69±1.27	78.99±2.14	2.38±0.22
Densenet [15]	89.50±2.85	82.65±2.24	1.51±0.75
Vnet [50]	89.70±3.51	82.94±2.93	1.38±0.32
TransBTS [40]	83.38±2.32	75.03±2.07	2.88±0.54
nnUnet [51]	89.71±2.68	82.87±2.15	2.30±0.61
our FTU3D	**90.53±1.36**	**84.14±1.92**	**1.27±0.45**

$$\text{Loss} = 1 - \frac{\sum_{i=1}^{N} w_i \cdot 2 \cdot |A_i \cap B_i|}{\sum_{i=1}^{N} w_i \cdot (|A_i| + |B_i|)} \tag{16}$$

Among them, N is the total number of classes, A_i and B_i correspond to the predicted and true segmentation for class i, and w_i represents the weight of class i. In our actual experiments, the number of classes is 2, with a weight of 0.9 for the segmentation target and 0.1 for the background target.

D. Experimental Details and Results

1) Comparison of Results Based on Artery Dataset: Intracranial artery segmentation is pivotal in medical image processing, seeking to accurately identify and segment intracranial arteries from raw CT angiography images. This task is vital for the diagnosis and treatment of brain diseases, including stroke, aneurysms, and vascular malformations. Due to our limited annotated data, we adopted a seven-fold cross-validation approach for training. The quantitative and qualitative evaluation results based on the Intracranial Artery dataset are presented in Table I and Fig. 2, respectively. Our model shows significant advantages in the accurate segmentation of intracranial arteries, reaching 90.53% and 84.14% in the Dice coefficient and IOU coefficient, respectively. Compared to CNN-based models, our proposed FTU3D demonstrated improvements in the Dice coefficient over nnUnet (0.82%), Vnet (0.83%), Densenet (1.03%), ResUnet (1.13%), and Unet (3.84%). In comparison to transformer-based models, our model exhibited enhancements in the Dice coefficient over SwinUNETR (0.74%), UNETR (2.38%), and TransBTS (7.15%). These significant performance advantage further underscore the efficacy of our model in handling complex anatomical structures. Moreover, we employed the ASSD metric to reflect the alignment precision of segmentation morphology by quantifying the average geometric distance between predicted segmentation and true segmentation. Our model achieves an average precision of 1.27 on this metric, showing improvements over SwinUNETR (34.19%), UNETR (35.85%), Unet (46.63%), and TransBTS (55.90%). These improvements prove that our method has also made outstanding contributions to improving the automated segmentation and reconstruction capabilities of human tissues with complex spatial relationships.

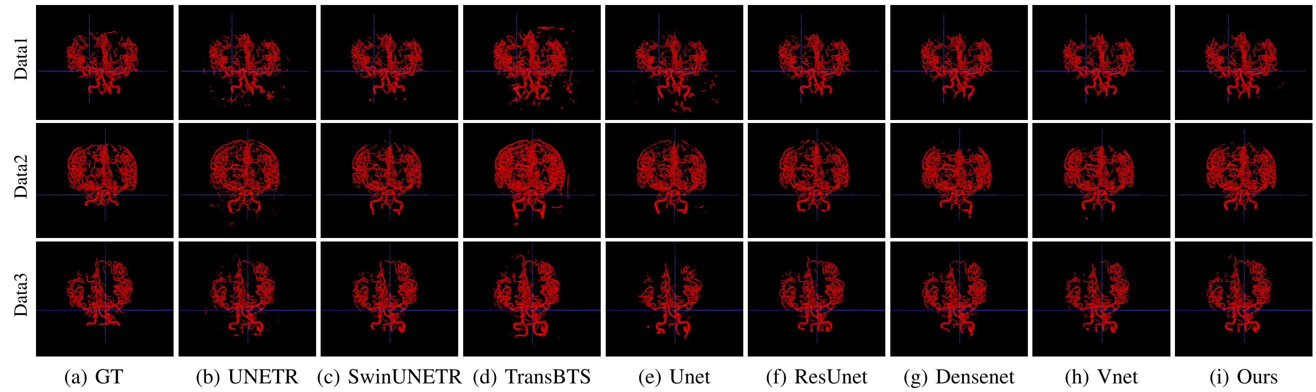


Fig. 2. Segmentation results using different models on intracranial artery dataset.

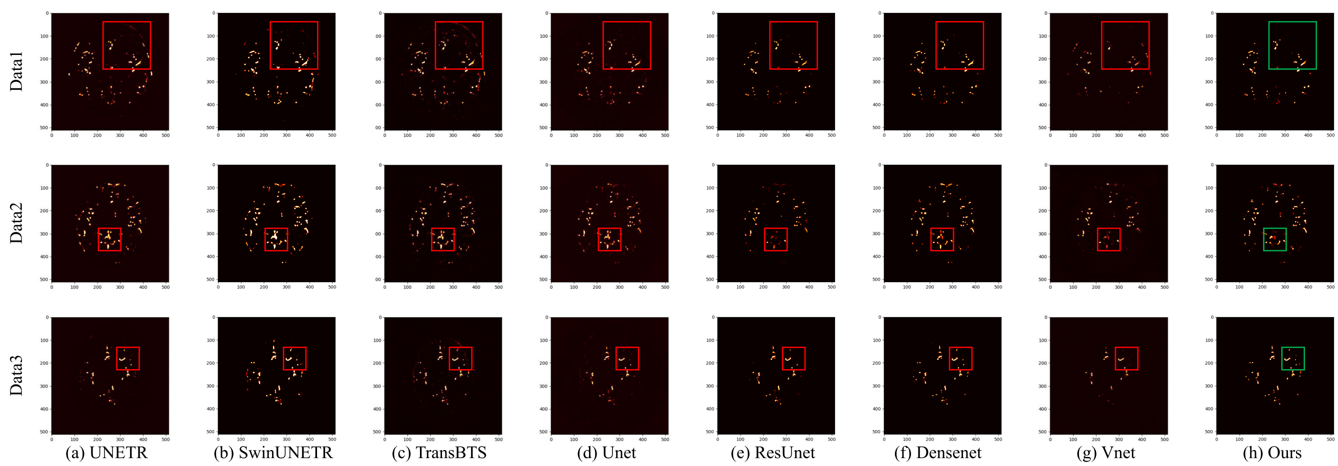


Fig. 3. Segmentation heatmap results using different models on intracranial artery dataset.

From the segmentation results in Fig. 2, we can see that our FTU3D model can maintain the continuity of the entire artery tissue and accurately reconstruct the spatial characteristics of the key regions where the aneurysm is located. In contrast, UNETR and TransBTS exhibited severe over-segmentation issues. The segmentation results of SwinUNETR faced problems of discontinuity and loss of details. Unet could only identify the more apparent structure of the major arteries, experiencing varying degrees of information loss and erroneous segmentation in smaller branch structures. Combined with the quantitative and qualitative evaluation results, our model accurately accomplished the segmentation reconstruction of the shape, size, and other aspects of the aneurysm-bearing artery tissue regions, providing an accurate data foundation for subsequent aneurysm segmentation.

As shown in Fig. 3, to enhance the clarity of the visual results, we added heatmap outputs to the test results in Fig. 2. From Fig. 3, it can be seen that transformer-based models incorrectly activated many background areas, leading to over-segmentation in segmentation results. CNN-based models exhibited insufficient activation in key feature regions, resulting in the omission of small arterial branch structures in the segmentation results. Furthermore, by observing the intensity distribution of the activation areas, it can be seen that the Densenet and SwinUNETR models generated intensely focused activations in certain regions of the input image. This caused these models to overemphasize arterial areas with specific structures, further explaining the overfitting issues present in these models. In contrast, our model demonstrated superior feature extraction and differentiation capabilities, effectively identifying and focusing on the actual lesion areas while giving minimal attention to background noise. This is the reason behind the performance improvement we ultimately achieved.

2) Comparison of Results Based on IA Dataset: The primary goal of aneurysm segmentation using CTA images is the precise segmentation and reconstruction of aneurysms, which is crucial for assessing rupture risk, tracking disease progression, and guiding surgical or interventional treatment strategies. The challenge is amplified by the small size of intracranial aneurysms and their substantial variability in size, shape, and location across patients.

We continue to employ a seven-fold cross-validation approach for training. The quantitative and some qualitative results based on the Intracranial Aneurysm dataset are presented in Table II and Fig. 4, respectively. Our model demonstrates superior performance, achieving Dice coefficient and IOU coefficient of 87.61% and 80.80%, respectively, and an average morphological precision of 6.11. The metrics indicate that our model can ensure precise segmentation of medical point targets and accurately reconstruct morphological

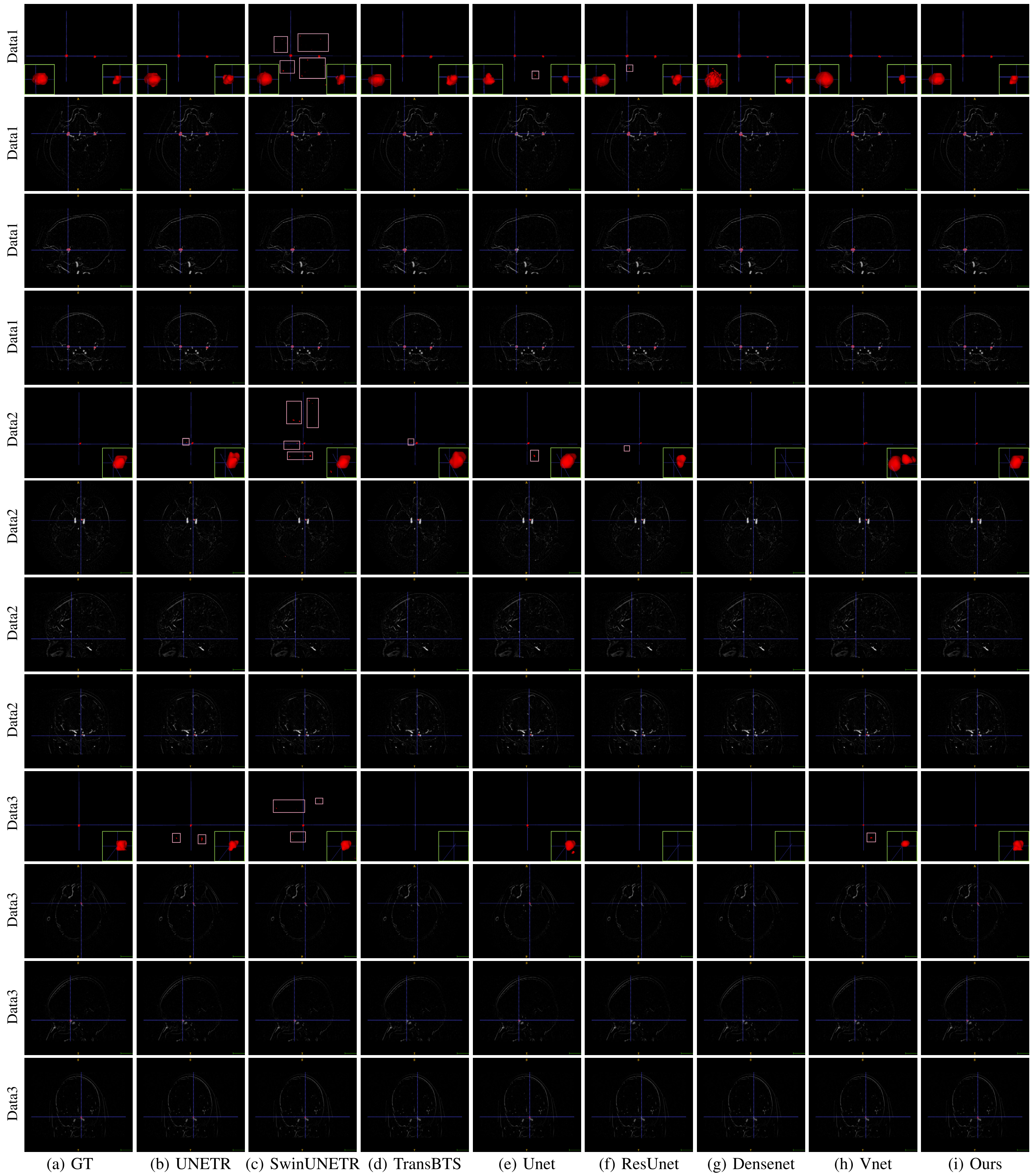


Fig. 4. Segmentation results using different models on intracranial aneurysm dataset.

information such as aneurysms. Compared to CNN-based models, our proposed model showed significant improvements in Dice coefficient over Unet (9.89%), nnUnet (18.70%), Vnet (21.07%), ResUnet (31.47%), and Densenet (35.38%). Compared to transformer-based models, our method also achieved significant enhancements in Dice coefficient over TransBTS (13.57%), UNETR (16.73%), and SwinUNETR (20.83%). These results highlight the comprehensive improvements in algorithm performance and usability of our FTU3D model for this difficult lesion segmentation task. More importantly, the FTU3D model successfully achieves a Dice coefficient of 87.61% for automatic segmentation algorithms in the intracranial aneurysm domain, marking a significant breakthrough that meets the 80% performance threshold for clinical applicability.

In terms of morphological accuracy, our model demonstrated a significant improvement of 89.78% and 73.56% over

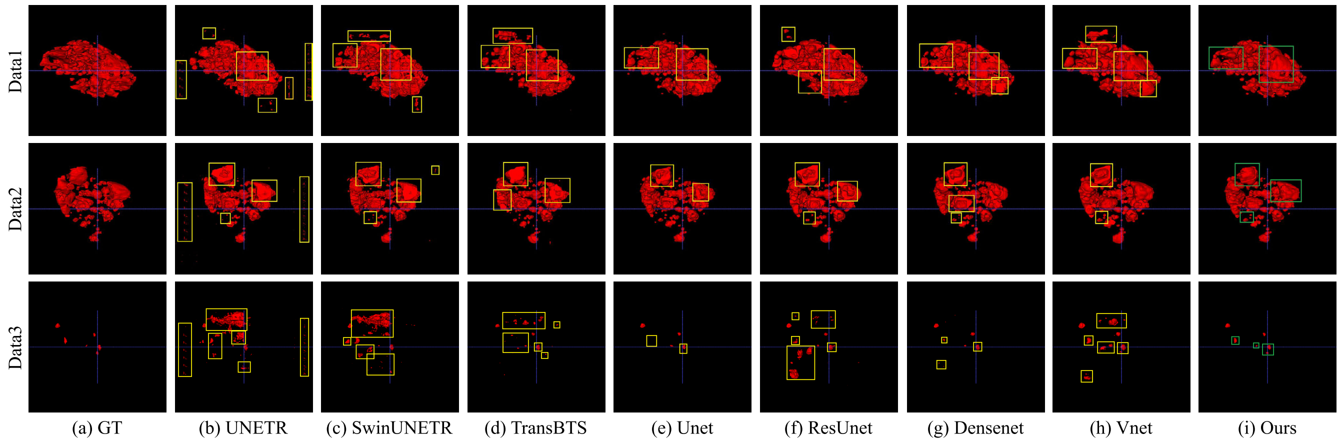


Fig. 5. Segmentation results using different models on LiTS17 dataset.

TABLE II
EXPERIMENTAL RESULTS BASED ON IA DATASET

Model	Dice(%)	IOU(%)	ASSD
UNETR [37]	70.88±3.64	64.69±2.82	INF
SwinUNETR [38]	66.78±2.01	60.61±1.45	59.81±8.97
ResUnet [36]	56.14±3.37	53.79±2.52	INF
Unet [14]	77.72±3.64	73.26±3.49	23.11±7.81
Densenet [15]	52.23±4.77	51.28±2.21	INF
Vnet [50]	66.54±2.35	60.82±1.97	INF
TransBTS [40]	74.04±5.23	66.91±3.35	INF
nnUnet [51]	68.91±4.81	62.14±3.62	INF
our FTU3D	**87.61±0.82**	**80.80±1.19**	**6.11±0.67**

TABLE III
EXPERIMENTAL RESULTS BASED ON LITS17 DATASET

Model	Dice(%)	IOU(%)	ASSD
UNETR [37]	79.70±1.75	74.38±1.64	39.46±4.45
SwinUNETR [38]	81.63±1.41	76.06±1.77	28.43±9.82
ResUnet [36]	80.59±1.09	75.19±0.81	19.15±4.77
Unet [14]	85.24±1.40	76.65±1.29	22.03±5.99
Densenet [15]	79.47±0.95	74.35±1.43	33.74±10.05
Vnet [50]	83.59±1.28	77.65±0.84	18.48±5.94
TransBTS [40]	82.35±1.05	76.29±0.79	26.74±5.43
nnUnet [51]	85.45±1.71	75.94±1.83	21.55±6.13
our FTU3D	**88.56±0.79**	**82.23±0.73**	**10.48±2.69**

Unet and SwinUNETR, respectively. It is important to note that, among all the existing models, aside from Unet and SwinUNETR, all other models exhibit cases of recognition failure. The cost of such failures in practical medical applications is substantial and deemed unacceptable. Many models perform well in intracranial artery segmentation but show significant performance degradation in aneurysm segmentation, such as Densenet and Vnet. While the limited quantity of annotated data is one of the reasons for the performance bottleneck of models, it is more due to these models' lack of proficient multi-scale segmentation capabilities. This further emphasizes the advanced nature of our model. The transition from artery segmentation to aneurysm segmentation involves distinctly different scenarios and target characteristics. Our model can overcome variations in target scale, shape, and spatial structure, showcasing strong generalization abilities that allow seamless adaptation to diverse medical scenarios.

We presented qualitative comparison results for the two stages of aneurysm segmentation in Fig. 2 and Fig. 4. From the outcomes of both stages, it's observable that our model achieves commendable overall performance. The performance differences among the models are further manifested in the three results of aneurysm segmentation we presented. Many models no longer meet the basic condition of a 100% detection rate. Our model not only completely recognized all aneurysm targets but also accurately retained the detailed information of the shape, position, and size of the aneurysms in each test sample. TransBTS, Densenet, and ResUnet failed to recognize aneurysms in Data2 and Data3, completely undermining their credibility as auxiliary diagnostic systems. SwinUNETR still had serious issues with over-segmentation. The test results of Vnet showed severe morphological deviations, making it unsuitable for subsequent processing.

Notably, the Data3 we presented is a challenging sample because the location of the aneurysm at the bend of the artery easily leads to missed detections and incomplete segmentation. Our model successfully identified and segmented it well, maintaining the basic morphological characteristics as seen in the figure. Unet and UNETR both experienced structural losses in their segmentation results for Data3, leading to morphological distortion. Only FTU3D exemplified our expectations for the role of deep learning technology. It can provide technical support in scenarios where human identification is challenging due to special positions or inconspicuous anomalies.

3) Comparison of Results Based on LiTS17 Dataset: To further enhance the contribution of the FTU3D model in more anticipated applications, we extended the comparative experiments to the liver tumor dataset by introducing the LiTS17 dataset for supplementary experiments. The LiTS17 dataset provides CT scan images from multiple centers and various devices, containing liver tumor data from different patients along with corresponding expert annotations. We continue to use the seven-fold cross-validation method for comparative experiments on the LiTS17 dataset. The quantitative and some qualitative results based on the LiTS17 dataset are presented in Table III and Fig. 5, respectively.

TABLE IV
RESULTS OF ABLATION EXPERIMENT

Num	Model	Cascade Fusion	Model Fusion Mode	Feature Fusion Mode	Dice
AE1	TFN3D only	×	×	×	0.5040
AE2	Unet3D only	×	×	×	0.7761
AE3	ResUnet3D only	×	×	×	0.5113
AE4	TFN3D with Unet3D	✓	direct fusion transmission	×	0.5968
AE5	TFN3D with Unet3D	✓	incremental fusion transmission	×	0.8657
AE6	TFN3D with ResUnet3D	✓	incremental fusion transmission	×	0.8057
AE7	Unet3D with TFN3D	✓	incremental fusion transmission	×	0.7025
AE8	TFN3D(2x) with Unet3D	✓	incremental fusion transmission	×	0.8661
AE9	TFN3Dv1 with Unet3D	✓	incremental fusion transmission	Add	0.8726
AE10	TFN3Dv2 with Unet3D	✓	incremental fusion transmission	HFF	**0.8747**

Our FTU3D model achieves a Dice coefficient of 88.56% and an IoU coefficient of 82.23%, significantly outperforming the other models. Compared to CNN-based models, FTU3D model showed significant improvements in Dice coefficient over nnUnet (3.11%), Unet (3.32%), Vnet (4.97%), ResUnet (7.97%), and Densenet (9.09%). Compared to transformer-based models, our model also achieved significant enhancements in Dice metric over TransBTS (6.21%), SwinUNETR (6.93%), and UNETR (8.86%). Moreover, in terms of ASSD metric, FTU3D model achieves an average precision of 10.48 on this metric, showing improvements over Vnet (43.29%), ResUnet (45.27%), nnUnet (51.36%), Unet (52.42%), TransBTS (60.80%), SwinUNETR (63.13%), Densenet (68.93%), and UNETR (73.44%). These performance advantages demonstrate that our model is not only capable of accurately segmenting medical point targets but also exhibits strong generalization in handling cases where the same lesion presents a wide range of volumetric variations. Our FTU3D model establishes a new benchmark for 3D liver tumor image segmentation, offering valuable insights for the development of more advanced models. This accomplishment highlights the broad applicability and high-precision segmentation capability of FTU3D across various medical scenarios, with significant implications for both clinical applications and academic research.

Some visualization test results based on the LiTS17 dataset are shown in Fig. 5. The figure illustrates the liver tumor lesions of different patients, which vary in size, shape, and distribution. In these various cases, our proposed model achieved commendable segmentation results and accurately segmented the morphology of the lesions. In contrast, other models exhibited numerous segmentation errors and omissions. These comparative models failed to accurately delineate the boundaries between lesions and the background when handling larger lesions, resulting in significant background noise in the segmentation results. When dealing with cases like Data3, which contains medical point targets within the overall lesion, these models showed incomplete recognition and even failures in identification. Such comparisons not only further prove the superior performance of our model but also demonstrate its broad applicability and capability to be extended to various medical 3D application scenarios.

4) Ablation Experiment: To validate the effectiveness of our design, we conducted ablation experiments based on the complete aneurysm segmentation task. The specific results are shown in Table IV. Among them, we designed a series of ablation experiments focusing on the incremental fusion architecture to verify its effectiveness and advantages. The results of experiments AE1, AE2, and AE3 indicate that using a single model makes it challenging to achieve high-precision segmentation of medical point target lesions, highlighting the necessity of model fusion. In the comparison of cascade fusion modes, The result of experiment AE4 shows that directly using the output of the previous model as the input for the subsequent model leads to performance degradation. This is because such a design increases the difficulty of model optimization, forcing the subsequent model to learn how to interpret the feature representations of the previous model, which can also result in uneven and unstable gradient transmission. The incremental fusion design in experiment AE5 significantly improved performance, ensuring that the model consistently maintains a direct connection with the original data. This encourages the preceding TFN3D model to focus more on identifying high-value regions. Consequently, the performance of AE5 improved by 36.17% and 15.70% compared to using single models in AE1 and AE2, respectively. Through comparative experiments, we found that ResUnet [36] performs better in intracranial artery segmentation than Unet [14]. Therefore, in AE6, we validated the practical effect of the ResUnet [36] model in the incremental fusion architecture. Although the actual effect is not as good as AE5, it has shown significant improvement compared to AE3. This demonstrates the adaptability of our proposed network structure, which can be extended to construct various combinations with other base models for mutual enhancement. Secondly, it indicates that the consideration of network complexity in AE5 is appropriate. Furthermore, the incremental fusion sequence experiment in AE7 demonstrates that the global feature extraction capability of the TFN3D model cannot be replaced by the convolutional neural network model, confirming the rationale of the FTU3D model design sequence. These experimental results comprehensively prove the effectiveness and scalability of the incremental fusion architecture in complex tasks.

In AE8, we attempted another downsampling rate. The result shows that increasing the input data resolution improves performance to some extent. However, it also brings more computational burden. We chose the original configuration ($r = 6$) based on the trade-off principle. In AE9 and AE10, we introduced different feature fusion strategies into TFN3D, namely direct addition and using the HFF module. The results

TABLE V
EXTERNAL TESTING RESULTS FOR INTRACRANIAL ANEURYSM AUXILIARY DIAGNOSIS

num	Model	Dice(%)	Dice pos(%)	Dice neg(%)	IOU(%)	Sensitivity(%)	Specificity(%)	ASSD
1	UNETR [37]	62.07	64.66	59.47	58.26	86.67(13/15)	60.00(9/15)	INF
2	SwinUNETR [38]	72.73	73.75	71.70	66.28	86.67(13/15)	86.67(13/15)	INF
3	Unet [14]	64.68	67.75	61.60	61.56	66.67(10/15)	60.00(9/15)	INF
4	TransBTS [40]	65.29	67.54	63.03	61.34	66.67(10/15)	46.67(7/15)	INF
5	our Focus-TransUnet3D	**84.14**	**80.79**	**87.49**	**77.95**	**100.00(15/15)**	**100.00(15/15)**	**8.81**

show that both strategies are effective. Compared with AE5, the segmentation performance has improvements of 0.69% and 0.90%, respectively. As hierarchical fusion is more suitable for the diverse scale characteristics of medical targets, we ultimately chose the HFF module. Overall, AE10 has achieved the best Dice accuracy currently, fully demonstrating the synergistic effect among the modules we designed and validating the correctness and advancement of our network architecture.

5) External Testing: We further conducted external tests on the capability of our model for auxiliary diagnosis of intracranial aneurysms in practical applications. By evaluating the model's performance on unseen clinical samples, we can ensure that the model not only performs well under laboratory conditions but also operates reliably in clinical practice. This external test included a total of 30 clinical samples, comprising 15 positive cases and 15 negative cases. Each sample data was the original CTA data, which was directly inputted into the deep learning model without any additional processing. The models selected for comparison only included those from the previous experiment that had a 100% detection rate or a Dice accuracy rate of over 70%. After two-stage automatic image segmentation, we summarized the actual effects of each model, displayed in Tab. V.

The test results reveal that the difficulty of external testing is incomparable to laboratory testing. Only our FTU3D passed the sensitivity and specificity tests with a 100% success rate, while other models encountered failures. Apart from SwinUNETR achieving 86.67% (13/15) in both sensitivity and specificity, and UNETR reaching 86.67% (13/15) in sensitivity, the accuracy of other models was below 80%. The specificity of TransBTS was even as low as 46.67% (7/15). This difference highlights the significant value of Focus-TransUnet3D in ensuring diagnostic precision, providing a strong guarantee against misdiagnosis and missed diagnosis. The Focus-TransUnet3D model also demonstrated significant advantages in key performance metrics. Specifically, the model achieved a Dice coefficient of 84.14%, showing a significant improvement compared to SwinUNETR (11.41%), TransBTS (18.85%), Unet (19.46%), and UNETR (22.07%). Similarly, in terms of IOU and ASSD, Focus-TransUnet3D led with results of 77.95% and 8.81, further validating its efficient performance in detail depiction and accuracy. This again demonstrates the outstanding performance of Focus-TransUnet3D in accurately delineating aneurysm boundaries and highlights its clear advantage in overall segmentation accuracy.

The excellent performance of our model is crucial for enhancing confidence in its deployment in practical applications. High sensitivity and high specificity enable clinicians to have full confidence in the practical utility of the model as a reliable diagnostic tool. Healthcare institutions can integrate such a model as a frontline screening tool into their diagnostic protocols to ensure high diagnostic accuracy and streamline patient management processes. In fact, this was also the decisive reason why our model was ultimately trialed in neurosurgery outpatient clinics. By providing highly accurate and reliable auxiliary diagnostic information, the model has the potential to improve the early detection and treatment planning of aneurysms. Our research not only demonstrates the capability of efficient deep learning methods in handling highly complex medical image data but also provides robust technical support for broader future clinical practice.

E. Discussion

1) Advantages and Practical Significance: Extensive experimental results demonstrate that our proposed method has successfully overcome the challenge of accurately segmenting intracranial aneurysms. While medical point targets are the primary focus of this study, our approach is not limited to lesion targets with this characteristic, nor are these characteristics necessary for the effectiveness of the proposed method. On the contrary, our model has achieved state-of-the-art performance across three different medical scenarios, accurately segmenting various irregularly shaped and complex medical targets, including intracranial artery segmentation, intracranial aneurysm segmentation, and liver tumor segmentation. These three tasks involve entirely different scenarios and target characteristics. Our proposed model can overcome variations in target scale, shape, and spatial structure, demonstrating strong generalization capabilities and seamlessly adapting to various medical scenarios.

More importantly, we have successfully applied the FTU3D model in clinical practice at the neurosurgery outpatient department, completing external testing of its auxiliary diagnostic capability for intracranial aneurysm patients. The practical auxiliary diagnostic performance of the FTU3D model has been recognized by neurosurgeons at Beijing Tiantan Hospital, demonstrating significant advantages in key performance metrics. The FTU3D model can accurately identify and segment intracranial aneurysm lesions from patient case images with a 100% success rate. The average Dice coefficient reached 84.14%, and the average IoU coefficient reached 77.95%. The ASSD metric, indicating morphological segmentation accuracy, achieved an average precision of 8.81. For such tiny and difficult-to-identify medical point targets, the FTU3D model not only performed well under labora-

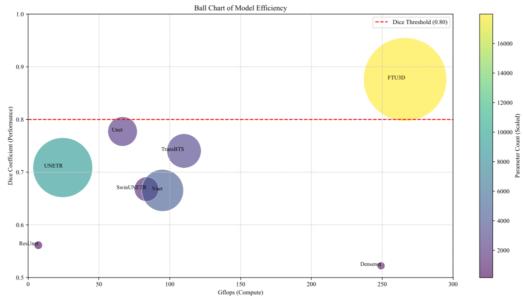


Fig. 6. Ball chart on segmentation performance (Dice) and model efficiency (Gflops) of different models.

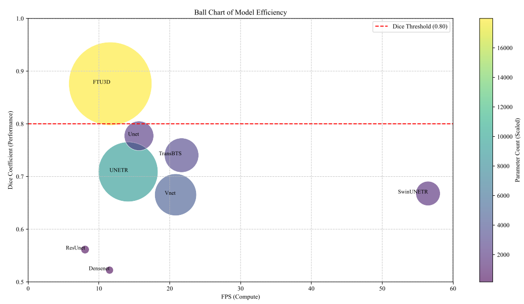


Fig. 7. Ball chart on segmentation performance (Dice) and model efficiency (FPS) of different models.

tory conditions but also reliably operated in clinical practice, highlighting its immense value and significant contributions to clinical applications. By providing highly accurate and reliable auxiliary diagnostic information, this model can effectively improve the quality of early detection of intracranial aneurysms, leading to better treatment outcomes for patients.

2) Limitations and Challenges: We established two ball charts to analyze the relationship between segmentation performance (Dice) and model efficiency for different models, as shown in Fig. 6 and 7. From the figures, it is evident that the FTU3D model has some optimization space in terms of image inference FPS and model Gflops. However, it is the only model that meets clinical application conditions (lesion segmentation Dice coefficient higher than 80%). Although from the current practical application, our model is fast enough for non-critical cases in neurosurgery outpatient settings. However, there is still a need to further improve its inference speed to expand its applicability, especially in acute care scenarios where faster decision-making is required. We are exploring various optimization techniques to reduce computation time without compromising the model's accuracy. Another area that needs improvement is the ability of our model to handle cross-modality data. Currently, the model has been mainly validated with a specific imaging modality (CT images), but its performance under other imaging modalities has not been thoroughly tested. Additionally, our current dataset is relatively small due to privacy concerns and the complexity of medical data annotation, making it challenging to obtain a large number of annotated 3D images. To address these issues, we plan to conduct further research to collect more imaging data from different modalities to increase the dataset size. This will also allow us to validate and optimize our model across different imaging types, such as MRI and PET scans, ensuring its robustness and adaptability to various clinical needs.

Through practical deployment, we have also identified some potential challenges, mainly related to the integration with existing medical data and information systems, as well as user acceptance issues. Looking ahead, we are concerned with the scalability of our auxiliary medical approach in different healthcare environments and its generalizability across diverse patient populations globally. To address these potential challenges, we will continuously update the model based on clinical data and feedback from medical professionals. Additionally, we are actively seeking opportunities to collaborate with medical centers worldwide to further promote our model. In our future research plans, we will also conduct larger-scale clinical trials to further verify the effectiveness of the model.

V. Conclusion

In this paper, we propose a high-performance deep learning model specifically designed for 3D medical point target segmentation, named Focus-TransUnet3D. This study represents a significant breakthrough in the field of aneurysm segmentation, overcoming challenges such as sparse annotations and tiny targets, to achieve precise segmentation of lesion areas at the point target scale. We enhance global feature extraction in 3D image segmentation model by incorporating a hierarchical 3D transformer architecture. A novel deep fusion strategy is also integrated, significantly improving the adaptability of our model to complex human structures and multi-scale targets. Addressing the challenge of limited training data for 3D deep learning methods in the medical field, we design an advanced incremental fusion architecture. Through incremental fusion optimization of feature transmission, our model achieves high performance with small-scale training data support. Multiple internal and external tests conducted on real clinical datasets validate the effectiveness of Focus-TransUnet3D. It demonstrates superior segmentation performance, achieving the best results on Artery dataset, IA dataset and LiTS17 dataset. However, there is still room for improvement in inference speed and multi-modality data support. In the future, we plan to extend the model to multi-class segmentation tasks under multi-modality conditions, continuing to promote the wide application of deep learning technology in the clinical field.

References

[1] L. Ma, G. Tan, H. Luo, Q. Liao, S. Li, and K. Li, "A novel deep learning framework for automatic recognition of thyroid gland and tissues of neck in ultrasound image," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 9, pp. 6113–6124, Sep. 2022.

[2] N. Gaggion, L. Mansilla, C. Mosquera, D. H. Milone, and E. Ferrante, "Improving anatomical plausibility in medical image segmentation via hybrid graph neural networks: Applications to chest X-ray analysis," *IEEE Trans. Med. Imag.*, vol. 42, no. 2, pp. 546–556, Feb. 2023.

[3] D. Xiang, B. Zhang, Y. Lu, and S. Deng, "Modality-specific segmentation network for lung tumor segmentation in PET-CT images," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 3, pp. 1237–1248, Mar. 2023.

[4] T. A. Soomro et al., "Image segmentation for MR brain tumor detection using machine learning: A review," *IEEE Rev. Biomed. Eng.*, vol. 16, pp. 70–90, 2023.

[5] Q. Zhao et al., "Efficient multi-organ segmentation from 3D abdominal CT images with lightweight network and knowledge distillation," *IEEE Trans. Med. Imag.*, vol. 42, no. 9, pp. 2513–2523, Sep. 2023.

[6] J. Dong, Y. Cong, G. Sun, Y. Yang, X. Xu, and Z. Ding, "Weakly-supervised cross-domain adaptation for endoscopic lesions segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 5, pp. 2020–2033, May 2021.

[7] F. Lin, Y. Xia, S. Song, N. Ravikumar, and A. F. Frangi, "High-throughput 3DRA segmentation of brain vasculature and aneurysms using deep learning," *Comput. Methods Programs Biomed.*, vol. 230, Mar. 2023, Art. no. 107355.

[8] Y. Yang, Y. Ju, and Z. Zhou, "A super lightweight and efficient SAR image ship detector," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[9] H. Huang et al., "UNet 3+: A full-scale connected UNet for medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 1055–1059.

[10] L. Zhang, K. Zhang, and H. Pan, "SUNet++: A deep network with channel attention for small-scale object segmentation on 3D medical images," *Tsinghua Sci. Technol.*, vol. 28, no. 4, pp. 628–638, Aug. 2023.

[11] L. Gu and X.-C. Cai, "Fusing 2D and 3D convolutional neural networks for the segmentation of aorta and coronary arteries from CT images," *Artif. Intell. Med.*, vol. 121, Nov. 2021, Art. no. 102189.

[12] Q.-Q. Chen, Z.-H. Sun, C.-F. Wei, E. Q. Wu, and D. Ming, "Semi-supervised 3D medical image segmentation based on dual-task consistent joint learning and task-level regularization," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 20, no. 4, pp. 2457–2467, Jul./Aug. 2023.

[13] H. Cui et al., "An improved combination of faster R-CNN and U-Net network for accurate multi-modality whole heart segmentation," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 7, pp. 3408–3419, Jul. 2023.

[14] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense volumetric segmentation from sparse annotation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Athens, Greece, 2016, pp. 424–432.

[15] T. D. Bui, J. Shin, and T. Moon, "3D densely convolutional networks for volumetric segmentation," 2017, *arXiv:1709.03199*.

[16] B. Li, "3D fully convolutional network for vehicle detection in point cloud," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2017, pp. 1513–1518.

[17] H. Lu, H. Wang, Q. Zhang, S. W. Yoon, and D. Won, "A 3D convolutional neural network for volumetric image semantic segmentation," *Proc. Manuf.*, vol. 39, pp. 422–428, Jan. 2019.

[18] J. Chen et al., "Medical image segmentation and reconstruction of prostate tumor based on 3D AlexNet," *Comput. Methods Programs Biomed.*, vol. 200, Mar. 2021, Art. no. 105878.

[19] M. Lin, Q. Cai, and J. Zhou, "3D Md-UNet: A novel model of multi-dataset collaboration for medical image segmentation," *Neurocomputing*, vol. 492, pp. 530–544, Jul. 2022.

[20] N. Alalwan, A. Abozeid, A. A. ElHabshy, and A. Alzahrani, "Efficient 3D deep learning model for medical image semantic segmentation," *Alexandria Eng. J.*, vol. 60, no. 1, pp. 1231–1239, Feb. 2021.

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Munich, Germany, 2015, pp. 234–241.

[22] A. Srivastava et al., "MSRF-Net: A multi-scale residual fusion network for biomedical image segmentation," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 5, pp. 2252–2263, May 2022.

[23] H. Li, Z. Yan, D.-H. Zhai, and Y. Xia, "ESNet: An efficient segmentation network for medical image segmentation," in *Proc. 42nd Chin. Control Conf. (CCC)*, Jul. 2023, pp. 8276–8281.

[24] D. Jha, M. A. Riegler, D. Johansen, P. Halvorsen, and H. D. Johansen, "DoubleU-Net: A deep convolutional neural network for medical image segmentation," in *Proc. IEEE 33rd Int. Symp. Comput.-Based Med. Syst. (CBMS)*, Jul. 2020, pp. 558–564.

[25] H. Li, D.-H. Zhai, and Y. Xia, "ERDUNet: An efficient residual double-coding UNet for medical image segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2083–2096, Apr. 2024.

[26] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[27] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[28] Y. Zhang, H. Liu, and Q. Hu, "Transfuse: Fusing transformers and CNNs for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Strasbourg, France, 2021, pp. 14–24.

[29] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 205–218.

[30] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[31] W. Shi, J. Xu, and P. Gao, "SSformer: A lightweight transformer for semantic segmentation," in *Proc. IEEE 24th Int. Workshop Multimedia Signal Process. (MMSP)*, Sep. 2022, pp. 1–5.

[32] J.-H. Shi, Q. Zhang, Y.-H. Tang, and Z.-Q. Zhang, "Polyp-mixer: An efficient context-aware MLP-based paradigm for polyp segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 1, pp. 30–42, Jan. 2023.

[33] R. Ge et al., "X-CTRSNet: 3D cervical vertebra CT reconstruction and segmentation directly from 2D X-ray images," *Knowl.-Based Syst.*, vol. 236, Jan. 2022, Art. no. 107680.

[34] G. Karayegen and M. F. Aksahin, "Brain tumor prediction on MR images with semantic segmentation by using deep learning network and 3D imaging of tumor region," *Biomed. Signal Process. Control*, vol. 66, Apr. 2021, Art. no. 102458.

[35] J. Cao et al., "2D–3D cascade network for glioma segmentation in multisequence MRI images using multiscale information," *Comput. Methods Programs Biomed.*, vol. 221, Jun. 2022, Art. no. 106894.

[36] H. Kataoka, T. Wakamiya, K. Hara, and Y. Satoh, "Would mega-scale datasets further enhance spatiotemporal 3D CNNs?" 2020, *arXiv:2004.04968*.

[37] A. Hatamizadeh et al., "UNETR: Transformers for 3D medical image segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 574–584.

[38] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *Proc. Int. MICCAI Brainlesion Workshop*, 2021, pp. 272–284.

[39] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[40] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, and J. Li, "TransBTS: Multimodal brain tumor segmentation using transformer," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Strasbourg, France, 2021, pp. 109–119.

[41] H. Cai, S. Li, L. Qi, Q. Yu, Y. Shi, and Y. Gao, "Orthogonal annotation benefits barely-supervised medical image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3302–3311.

[42] R. A. Sinnaswamy, N. Palanisamy, K. Subramaniam, S. Muthusamy, R. Lamba, and S. Sekaran, "An extensive review on deep learning and machine learning intervention in prediction and classification of types of aneurysms," *Wireless Pers. Commun.*, vol. 131, no. 3, pp. 2055–2080, Aug. 2023.

[43] Y. Zeng et al., "Automatic diagnosis based on spatial information fusion feature for intracranial aneurysm," *IEEE Trans. Med. Imag.*, vol. 39, no. 5, pp. 1448–1458, May 2020.

[44] R. Shahzad et al., "Fully automated detection and segmentation of intracranial aneurysms in subarachnoid hemorrhage on CTA using deep learning," *Sci. Rep.*, vol. 10, no. 1, p. 21799, Dec. 2020.

[45] W. Yuan, Y. Peng, Y. Guo, Y. Ren, and Q. Xue, "DCAU-Net: Dense convolutional attention U-Net for segmentation of intracranial aneurysm images," *Vis. Comput. Ind., Biomed., Art*, vol. 5, no. 1, pp. 1–18, Dec. 2022.

[46] C. Ou et al., "A deep learning–based automatic system for intracranial aneurysms diagnosis on three-dimensional digital subtraction angiographic images," *Med. Phys.*, vol. 49, no. 11, pp. 7038–7053, Nov. 2022.

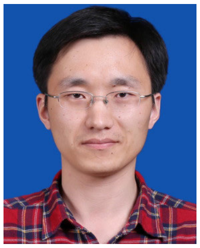
[47] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[48] H. Wang, X. Hu, X. Zhao, and Y. Zhang, "Wide weighted attention multi-scale network for accurate MR image super-resolution," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 3, pp. 962–975, Mar. 2022.

[49] P. Bilic et al., "The liver tumor segmentation benchmark (LiTS)," *Med. Image Anal.*, vol. 84, Apr. 2023, Art. no. 102680.

[50] F. Milletari, N. Navab, and S. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[51] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, no. 2, pp. 203–211, Feb. 2021.



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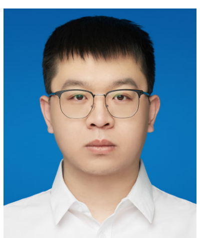

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